Day 4 Evidence Workbooklet — Instructor Key

Comparisons and Boundary Cases

Engineering practice → threshold rule → Python comparison → boundary evidence → support route

Instructor key. Same layout as student copy; solutions filled in response boxes. Print format: duplex, left-stapled packet.

Name		**Section**		**Date**	
Score	______/44		Mastery check	secure / developing / needs support	

Yesterday the pump-flow intake values were converted from text into numeric fields. Today the team uses those typed values to check a quality boundary: a flow reading is acceptable if it is inside a target band. The exact boundary rows matter because a reading exactly at the lower or upper limit may be allowed depending on the operator.

The problem today is comparison evidence: below, at, or above a limit; inclusive versus exclusive rules; and an exact sentence that explains a boundary case.

Engineering task	Check whether converted pump-flow readings meet an acceptable range before the team uses them in a go/no-go gate.
Computational move	Evaluate one comparison at a time and explain exact-boundary cases without jumping ahead to compound decisions.
Python focus	`<`, `<=`, `>`, `>=`, `==`, `!=`, comparison result, inclusive limit, exclusive limit.
Evidence to keep	reading value, boundary value, operator, true/false result, and a sentence explaining whether the boundary is included.

Day 4 boundary rules

1. A comparison expression produces `True` or `False`.
2. `<=` and `>=` include equality at the boundary.
3. `<` and `>` do not include equality at the boundary.
4. `==` checks equality; it does not assign a value.
5. A reliable threshold note names the value, the limit, the operator, and the result.

1 Read the threshold notebook one comparison at a time

recognize predict

Use the converted numeric fields from the intake record. Do not combine the comparisons yet; Day 5 will do that.

```
# Converted numeric fields from the intake record
target_flow_lpm = 20.0
low_limit_lpm = 19.6
high_limit_lpm = 20.4
reading_low_edge = 19.6
reading_middle = 20.0
reading_high_edge = 20.4
reading_high_miss = 20.5

# Individual comparison checks
reading_low_edge >= low_limit_lpm
reading_low_edge > low_limit_lpm
reading_high_edge <= high_limit_lpm
reading_high_edge < high_limit_lpm
reading_high_miss <= high_limit_lpm
reading_middle == target_flow_lpm
```

Pts.	Prompt	My evidence
1	Which variables define the lower and upper allowed flow limits?	**Key:** `low_limit_lpm` and `high_limit_lpm`.
1	Which reading is exactly equal to the lower limit?	**Key:** `reading_low_edge` = 19.6.
1	Which reading is exactly equal to the upper limit?	**Key:** `reading_high_edge` = 20.4.
1	Which reading is above the upper limit?	**Key:** `reading_high_miss` = 20.5.

2 Predict comparison results at exact boundaries

predict explain

Complete the table. Explain the result using the operator, not just the number.

Pts.	Comparison expression	True or False?	Boundary reason
2	`reading_low_edge >= low_limit_lpm`	**Key:** `True`	**Key:** Equal to lower limit; `>=` includes equality.
2	`reading_low_edge > low_limit_lpm`	**Key:** `False`	**Key:** Equal is not greater than.
2	`reading_high_edge <= high_limit_lpm`	**Key:** `True`	**Key:** Equal to upper limit; `<=` includes equality.
2	`reading_high_edge < high_limit_lpm`	**Key:** `False`	**Key:** Equal is not less than.

Common trap to check

A reading exactly at the limit is included by `<=` or `>=`, but it is not included by `<` or `>`. The operator decides the boundary, not the word "limit" by itself.

3 Separate equality from assignment and mismatch checks

recognize | predict

Python uses `==` for equality checks and `=` for assignment. This page keeps comparison evidence separate from stored state.

Pts.	Expression or line	Result or effect	What evidence does it create?
2	`reading_middle == target_flow_lpm`	**Key:** `True`	**Key:** Equality comparison; no assignment.
2	`reading_high_miss != target_flow_lpm`	**Key:** `True`	**Key:** `20.5` is not equal to `20.0`.
2	`reading_middle = target_flow_lpm`	**Key:** assignment	**Key:** Stores `20.0` in `reading_middle`; not a comparison.
2	`reading_high_miss <= high_limit_lpm`	**Key:** `False`	**Key:** `20.5` is above `20.4`.

Pts.	Prompt	My response
2	Why is `==` a safer symbol than the word "is" when writing comparison evidence?	**Key:** `==` names an equality comparison explicitly; “is” can be ambiguous in prose.
2	Why should a threshold note not say only "near the target"?	**Key:** It must name the value, limit, operator, and whether equality is included.

4 Build a boundary table before combining conditions

trace explain

Complete the boundary table. The last column should be a short engineering sentence, not only `True` or `False`.

Pts.	Reading	Lower check: `>=` `19.6`	Upper check: `<=` `20.4`	Boundary evidence sentence
2	`19.5`	**Key:** False	**Key:** True	**Key:** Below lower limit; not allowed.
2	`19.6`	**Key:** True	**Key:** True	**Key:** At lower limit; inclusive lower boundary allows it.
2	`20.4`	**Key:** True	**Key:** True	**Key:** At upper limit; inclusive upper boundary allows it.
2	`20.5`	**Key:** True	**Key:** False	**Key:** Above upper limit; not allowed.

Bridge to Day 5

Tomorrow the team will combine separate comparison results into a go/no-go expression. Today the important habit is to make each comparison traceable first: value, operator, limit, result, and boundary sentence.

5 Debug a boundary claim

debug test explain

A teammate writes this note after checking the lower edge reading:

Boundary note to debug

The `19.6` L/min reading fails because it is not greater than the lower limit of `19.6` L/min.

Pts.	Prompt	My response
2	What part of the teammate's sentence uses a real comparison result?	**Key:** `19.6 > 19.6` is `False`.
2	What rule did the teammate use that may not match the allowed boundary?	**Key:** They used an exclusive `>` rule. If the allowed boundary is inclusive, use `>=`.
2	Write the inclusive comparison that would allow the exact lower-limit reading.	**Key:** `reading_low_edge >= low_limit_lpm`.
2	Rewrite the boundary note so it names the operator and explains whether the exact limit is included.	**Key:** Using `>=`, `19.6 >= 19.6` is `True`, so the exact lower-limit reading is allowed.

Boundary-test habit

When a threshold rule matters, test at least three rows: just below the boundary, exactly at the boundary, and just above the boundary. Exact-boundary rows catch many comparison mistakes.

6 Mastery check and support route

transfer explain

Use your own evidence from this workbooklet. Do not write a generic answer.

Pts.	Prompt	My response
2	Give one example where changing from `>` to `>=` changes the result at an exact boundary.	**Key:** `reading_low_edge > low_limit_lpm` is False, but `reading_low_edge >= low_limit_lpm` is True.
2	In one or two sentences, explain why Day 3 type/cast evidence is needed before Day 4 boundary checks.	**Key:** Boundary comparisons require numeric values. Day 3 casting makes sure text fields have become floats before comparison.

My mastery check

Mark the strongest statement that is true after this workbooklet.

☐	Secure: I can evaluate comparison operators and explain exact lower or upper boundary cases.
☐	Developing: I can evaluate some comparisons, but I still need support with inclusive versus exclusive limits.
☐	Needs support: I need a smaller example before I can trust my boundary-check evidence.

My next support move

My issue is mostly: setup / Python syntax / tracing / operator choice / boundary cases / confidence / time.
The first boundary where I got uncertain was: ________________________________
My next small action is: ask a partner / ask instructor / rebuild the boundary table / test just-below-at-above rows / try the recovery version.

Workbooklet target: I can evaluate Python comparison operators, explain exact-boundary cases, and prepare separate comparison evidence before Boolean combinations.